

ADAM MORRIS

Engineering Mathematics PhD Researcher | Robotics & Autonomous Systems

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PROFILE

Engineering Mathematics PhD researcher investigating robot swarms for RF monitoring and concealed-emitter localisation, with relevance to defence and security. Combines simulation and robotic platforms with experience in AI consultancy, technical reporting and student supervision.

EDUCATION & RESEARCH

PhD in Engineering Mathematics | University of Bristol

2024 - 2028 (expected)

Thesis: Swarm-Enabled Spatial Intelligence

- Investigating how autonomous robot teams can monitor and maintain the RF domain in secure environments, including localising anomalous signals from concealed adversarial transmitters.
- Using 2D simulations for swarm design, high-fidelity 3D electromagnetic propagation engines and ROS 2/Gazebo for robot-control simulation.
- Working with physical robotic platforms, including the LEO Rover and Clearpath Jackal.

MEng in Engineering Mathematics | University of Bristol

2020 - 2024

First Class Honours | **Dissertation:** Anti-aligning robots

EXPERIENCE

Consultant Engineer | SportsViz

June 2023 - present

Part-time, remote

- Research AI and data interpretation for elite sports performance analysis in a project with the Football Association of Wales (FAW).
- Authored literature reviews and technical reports outlining future research directions.
- Demonstrated AI and data analysis capabilities for international-level sport to clients from varied professional backgrounds.

Graduate Teaching Support | University of Bristol

September 2023 - present

- Support undergraduates and master's students with technical writing, research methodologies and marking; provide one-to-one master's dissertation supervision.
- Rapidly learn unfamiliar topics to support student work and coordinate meetings across multiple groups.

TECHNICAL SKILLS & LANGUAGES

Proficient: Python, MATLAB, LaTeX, ROS 2

Experienced: C/C++, Arduino

Simulation & platforms: Gazebo, 2D swarm simulation, 3D EM propagation modelling, LEO Rover, Clearpath Jackal

Languages: English and Welsh (fluent)

PUBLICATION

Adam Morris, Timothy Pelham and Edmund R. Hunt. Spatially Intelligent Patrol Routes for Concealed Emitter Localization by Robot Swarms. IEEE SSRR 2025, Galway, Ireland, pp. 108-114. [IEEE Xplore](#)

SELECTED PROJECTS

[Project title] - [Your contribution, tools used and key result.]

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